



# A computer graphics-based model to generate dynamic 3D animations for corresponding Bangla sign language gestures using HamNoSys to SiGML conversion

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## Abstract

Effective communication is essential for human touch, and it's especially crucial for those who rely on sign language because they have speech or hearing difficulties. Communication between deaf and normal people is restricted by the inability of current technology to automatically create flexible Bangla Sign Language (BdSL) animations from Bangla text or voice. This work introduces a revolutionary computer graphics-based system that takes voice or text input in Bangla and uses it to create BdSL animations. The system uses a HamNoSys to Gesture Markup Language (SiGML) conversion to dynamically translate input into gestures. With the use of 94 classes in the dataset that cover Bangla numbers, alphabets, and word motions, the model can generate any word or phrase in Bangla. Additionally, the system creates gestures for inputs that aren't in the dataset by spelling out the letters. The system converts Bangla text into three-dimensional animated BdSL movements by parsing it based on Linguistic principles. According to performance evaluations, each input has a processing cost of 79.57 ms, and the average accuracy for text and voice input is 97.50% and 94.75%, respectively. The method ensures fluency and naturalness by taking into consideration crucial elements of sign language, such as hand shape, palm orientation, and non-manual messages. By reducing communication barriers between the sign and non-sign populations, this study significantly advances accessibility. Visit this GitHub repository link for further details on the implementation of the suggested system and the SiGML dataset: <https://gitlab.com/devarifkhan/bdsl-3d-animation>

**Keywords** Computer graphics · Sign language · Sign markup language · Deaf communication · Hamburg notations · Machine translation · Datasets · BdSL · SiGML · Hamnosys

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## 1 Introduction

In human civilization, survival is difficult without communication. There are several ways for people to communicate and share ideas. The two most common ways to communicate are by speaking and using gestures. Through vocal or audible sounds, the sense of hearing is employed to transmit the language of speech. People also utilize gestures to express their views, notably arm motions over different body parts (Caballero-Morales and Trujillo-Romero 2013; Chowdhury et al. 2024). Deaf people frequently communicate manually by using sign language. Individuals with hearing loss can communicate more easily using sign language. It serves as an effective and visual medium for expressing ideas, sentiments, and feelings. However, few thorough materials are available for sign language, especially in lesser-used languages like Bangla. Deaf people worldwide use a variety of sign languages; there is no single sign language used by all Deaf people worldwide. Depending on the geographic area, many spoken and sign languages are utilised (Brouer and Benabbou 2021). The second most frequently spoken among India's 22 scheduled languages, Bangla is the official national language of Bangladesh and the language that is spoken there the most. Bengali is the world's fifth-most-spoken native language and the seventh-most-spoken language overall, with around 300 million native speakers and an additional 37 million people who speak it as a second language. For the Deaf people in Bangladesh, the Bangla Sign Language was their major language of exchange. Not many individuals are familiar with sign language (Caballero-Morales and Trujillo-Romero 2013).

For Bangladesh's Deaf community, BdSL is a vital way of communicating. About 2.6 million people in Bangladesh are unable to communicate with others using language (Shamrat et al. 2021; Islam et al. 2021). A natural sign language used by the Deaf community in Bangladesh is called BdSL. The linguistic characteristics of BdSL are crucial, and promoting communication among the sign population is hard. The need for additional study and funding to assist in the development and maintenance of BdSL is important (Khatun et al. 2021). However, there is still a communication gap between normal people and deaf people, and they are unable to grasp sign language (Ahmed et al. 2016). Even in regions where spoken languages are similar, sign languages vary in terms of syntax, vocabulary, and usage of signs. This is because sign languages are influenced by different nations and cultures (Caballero-Morales and Trujillo-Romero 2013). The lack of a Bangla sign language dataset may be the primary cause of the research gap (Islalm et al. 2019). Concatenating sign words that were collected separately allows for the realization of sign language animation depending on a key characteristic. This indicates that a word in sign language follows the same pattern in a variety of contexts, which contrasts with realistic sign language expression (Li et al. 2014). Particularly in Bangladesh, where a lack of resources prevents individuals from learning the BdSL. Few people know sign language, so integrating them into society is difficult. The reality is that only around 5% of people worldwide can communicate using sign language (Hassan 2022). On the globe, there are more than 200 distinct sign languages. Researchers have created sign language libraries that include gestures to make it easier for Deaf people to learn sign languages (Farooq et al. 2021). The availability of language resources, standardisation of notation systems, and modelling realistic and emotive facial expressions all pose difficulties for creating BdSL motions. This study offers a novel method for automatically creating animated Bangla sign language gestures from Bangla text and voice to address this problem. 3D animated sign language has practical applications in a

variety of domains, including sign language research projects in Bangla, educational tools, interpretation and avatars, communication tools for the deaf, public services and broadcasting, healthcare communication, and legal and administrative services. These apps have enormous potential to improve communication, education, and accessibility for sign language users, hence simplifying their incorporation into various aspects of everyday Life. We hope to build inclusive settings and empower individuals by using 3D HamNoSys to boost participation and engagement chances.

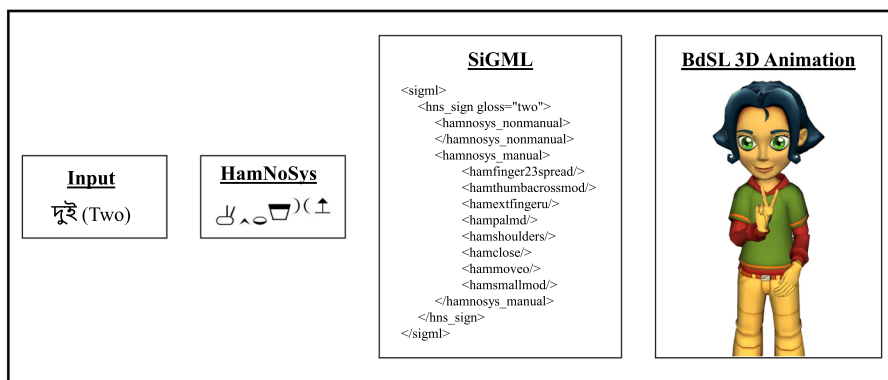
In addition to exposing the larger socioeconomic issues connected with sign language generation and communication hurdles, it is critical to investigate the specific scientific complexity underpinning the construction of 3D avatars for sign language. Accurate and expressive sign language animations require solving various technological challenges, including capturing the fine intricacies of hand movements, hand shape, palm orientation, location, and expression/non-manual signals. To ensure that these avatars are realistic and Lifelike, advanced motion capture, animation synthesis, and rendering methods are required. Furthermore, attaining flawless synchronisation of language components and visual representations involves a distinct set of obstacles. By tackling these specific scientific issues, we want to contribute to the evolution of technology and promote successful communication for those who use sign language. We highlight the importance of our work within the field of computer graphics and visual techniques approaches. Our proposed model creates excellent 3D gesture animations in Bangla Sign Language by utilizing cutting-edge technologies. This study offers creative approaches to precise and Lifelike gesture representation, which advances the larger objectives of computer animation and virtual environments. Our approach, based on converting HamNoSys to SiGML, presents a new use case for graphics-oriented technologies. Our research introduces a revolutionary system that uses text and voice inputs to produce 3D gesture animations in Bangla Sign Language (BdSL), which significantly advances the area of computer graphics.

Gestures and symbols have a grammatical structure in sign language. The three main elements of each sign are the hands' shape, movement, and position. Similar to spoken languages, this intricate communication system enables the expression of complex thoughts and emotions. But most people who are not Deaf themselves are not familiar with sign language. This makes it extremely difficult for hearing people and sign language users to communicate. These organizations usually need a translator to translate spoken words into sign language and vice versa during conversations. The hands of the signer are the main component of sign language. Sign language gestures are distinguished by certain hand shapes, orientations, postures, and motions. Together, these components provide a visually striking and expressive way to communicate meaning. A viable method for displaying sign language understandably and straightforwardly is to use dynamic 3D animations. These animations can accurately depict the complex hand gestures and postures necessary for clear sign language communication. We have suggested a 3D animated system to convert voice and text in Bangla into Bangla Sign Language effectively. The goal of this project is to improve automated systems for creating sign language by focusing on Bangla Sign Language. The suggested approach is taking the extensive system for transcribing sign language motions, known as HamNoSys (Hamburg Notation System) (Hanke 2004), and translating them into SiGML (Signing Gesture Markup Language) (University of Hamburg. Sigml 2025). With HamNoSys, sign language motions may be accurately and thoroughly described. SiGML can then be used to convert these descriptions into 3D animations. Our 3D animated system

uses voice and text input in Bangla to generate realistic and accurate animated sign language movements. This system creates realistic and dynamic representations of Bangla Sign Language motions by utilizing state-of-the-art computer graphics and animation techniques.

Figure 1 shows the basic idea of the proposed system. Numerous accessibility issues affect the Deaf community's quality of life and ability to integrate into society. Since interactions frequently include spoken and written language, which sign language users may find inaccessible, communication obstacles in jobs, education, and healthcare are the main causes of worry. These obstacles cause Deaf people to have worse quality healthcare, greater unemployment rates, and disadvantages in school. Mental health and general well-being are also impacted by social isolation brought on by communication problems. The research aims to improve accessibility and communication for the deaf community by employing the Bangla sign language. This study creates a comprehensive SiGML data class of Bangla sign language gestures, creating a system for automatically generating gestures and improving communication and accessibility for Bangla-speaking people with hearing challenges. Its findings might result in the development of tools and systems that automate the production of sign language movements from Bangla text and voice inputs, thereby facilitating inclusive communication. Additionally, the study can support accessibility in public areas, entertainment media, and real-time translation, promoting inclusion in various fields and assisting Deaf people. The following contributions are made by us—

- To the best of our knowledge, the proposed system is the first complete BdSL animation system for converting Bangla text or voice into 3D BdSL animated gestures.
- We developed a rule-based parser for producing BdSL based on Bangla linguistics and built a SiGML dataset using the Hamburg Notation System (HamNoSys) to SiGML.
- We present a model that can produce Bangla Sign Language gestures on its own by using the HamNoSys to SiGML conversion and the Bangla linguistic feature of BdSL. It has a spelling system built in; it can dynamically generate any Bangla word, sentence, and composite numeral gestures based on the trained HamnoSys to SiGML dataset instead of training individual numerals, alphabets, and words.
- Bangla is a very complicated language, with substantial regional differences in pronunciation, vocabulary, and syntax. To solve these issues, our system has spelling skills that enable it to handle complicated phrases and words by splitting them down into



**Fig. 1** Procedure for converting text into animated sign

constituent letters. This feature guarantees that even if a word or phrase is not explicitly included in the dataset, the algorithm can produce appropriate BdSL motions by spelling out the letters.

- The communication accessibility for deaf and normal people will be significantly improved by introducing our proposed automatic gesture animation generation system for BdSL. The technique can be used as a teaching aid for Bangla Sign Language. More individuals will be motivated to learn and utilize sign language if interactive and entertaining 3D animations make the process more fun and straightforward.

The rest of the work is methodically organized to facilitate a thorough investigation. Existing systems are critically evaluated in the Literature Review in Sect. 2. Most importantly, the system's design is explained under the Proposed Methodology in Sect. 3. The Result Analysis with our large dataset of the experiments is described in Sect. 4. Finally, the paper is concluded in Sect. 5, with Limitations, applications, and future research. The scholarly foundation of the text is emphasised through citations and references.

Table 1 contains a list of the terms and definitions used in this essay.

## 2 Literature review

The automatic generation of sign language has received a lot of interest lately because of its potential to improve communication between sign language users and non-users. The study on automatically creating animated Bangla sign language gestures from Bangla text and speech is covered in detail in this survey of the Literature. Not much progress has been made in translating sign language from text or voice into 3D animation, despite a large body of research on sign language prediction and recognition (Tyagi and Bansal 2020). The difficulties involved in creating Bangla Sign Language are particularly great, as it is one of the most complex sign languages in the world. This is demonstrated by the fact that there are 49

**Table 1** Terminologies are listed alphabetically

| Terms    | Description                                         |
|----------|-----------------------------------------------------|
| 3D       | Three Dimensional                                   |
| SL       | Sign Language                                       |
| BdSL     | Bangla Sign Language                                |
| ISL      | Indian Sign Language                                |
| PSL      | Pakistani Sign Language                             |
| ASL      | American Sign Language                              |
| ARSL     | Arabic Sign Language                                |
| XML      | eXtensible Markup Language                          |
| SiGML    | Signing Gesture Markup Language                     |
| HamNoSys | Hamburg Notation System                             |
| GRS-BdSL | Gesture Recognition System for Bangla Sign Language |
| API      | Application Programming Interface                   |
| SLR      | Sign Language Recognition                           |
| TVML     | TV program Making Language                          |
| CG       | Computer Graphics                                   |
| CNN      | Convolutional Neural Network                        |
| IDE      | Integrated Development Environment                  |
| GUI      | Graphical User Interface                            |

different sign language styles for only the Bengali alphabet (Hasib et al. 2023). The creation of signing avatars, as demonstrated by the efforts of Rahaman et al. and P. C. Badhe et al., is a noteworthy addition. These studies also introduce novel technology for dynamic Bangla sign motions (Islam et al. 2022; Badhe and Kulkarni 2015). Rahaman et al. concentrate on the development of Bangla numerical sign language animations, which are capable of producing any type of Bangla composite number. Additionally, P. C. Badhe et al. employ the same method to generate Indian sign language. These studies focus on techniques and innovations that provide advantages and disadvantages as they investigate current advancements in the field of animated sign language creation. However, our suggested system is a comprehensive set of gesture production tools for Bangla sign language that can produce any Bangla letter, number, word, and phrase

The literature on different methods for sign language technology is critically reviewed in this part, with an emphasis on the methods' advantages, disadvantages, and cutting-edge characteristics. Convolutional Neural Networks (CNNs) and Graph Neural Networks (GNNs) are two recent approaches to sign language technology. According to "A Comprehensive Review of CNN-Based Sign Language Translation System," CNNs are good at identifying both static and dynamic motions, but they have trouble understanding the temporal features of continuous signing (Seema and Singla 2023). HST-GNNs, which simulate intricate interactions between body parts for more accurate translations, are introduced in "Sign Language Translation with Hierarchical Spatio-Temporal Graph Neural Network" (Kan et al. 2022). Although this approach needs comprehensive annotations and has greater computing costs, it excels at capturing the subtleties of sign language. While both methods improve sign language recognition, GNNs are more complicated and precise. Also, sign language recognition and translation systems have some shortcomings. Rahaman et al. substantially contributed to addressing the shortcomings of SLR systems by introducing a six-layered convolutional neural network for effective identification of Bangla sign language, reaching outstanding accuracy rates of 92.78% and 98.38% (Rahaman et al. 2023). Kipp et al. addressed the difficulties in making a signing avatar and enhanced the EMBR character animation system through professional sign language assistance, the addition of a gloss-based animation tool, and the creation of a cyclical process for content generation (Kipp et al. 2011).

Using attention strength and neutral state data from a sign language instructor, Li Jinghua et al. developed a computer model that presented a novel method for producing gestures in sign language (Jinghua et al. 2012). A Microsoft Kinect RGB-D camera and computer vision algorithms were used by Bhagat et al. (2019) to develop a real-time hand gesture detection system. In parallel, a dynamic hand gesture identification technique utilizing deep learning architectures was developed by Al-Hammadi et al. (2020). Protik is an innovative multimedia teaching tool for kids with hearing problems that is being used to teach Bangla Sign Language (BdSL) (Nawshin et al. 2020). Protik improves the learning process by using captivating animations and graphical depictions that help young learners comprehend and mimic motions more effectively. Protik and our study both highlight the significance of visual and interactive technology in enhancing Bangla sign language instruction, and the incorporation of HamNoSys to SiGML conversion makes it possible to create precise and dynamic 3D gesture representations. Liang et al. created an interactive puppetry system that utilizes hand gestures in a relevant context, demonstrating the potential of gesture-based systems to enhance children's storytelling in educational environments (Liang et al. 2017).

Specifically, our suggested system can convert HamNoSys to SiGML and generate dynamic 3D animations, which sets it apart. This method not only increases the accuracy of gesture representations but also broadens and intensifies the Bangla Sign Language learning experience. Comparing our model to past efforts, it offers a unique mix of high accuracy, flexible generating capabilities, and significant educational value by filling in the gaps found in the literature.

Recent breakthroughs in sign language production have harnessed deep learning and Natural Language Processing (NLP) to increase communication for the Deaf and hard-of-hearing groups. By using methods like transformers and sequence-to-sequence structures, sign language may now be automatically generated from written or spoken inputs with great success. Nevertheless, there are still issues with real-time creation, fine-grained gesture identification, and accounting for variances among sign languages. To ensure coherence and naturalness in generated gestures, a major focus has been on creating vast, diverse datasets. Additionally, ethical aspects such as cultural sensitivity and user permission have been addressed (Techniques for Generating Sign Language: A Comprehensive Review) (Waghmare 2024). The Structure-aware Temporal Consistency Network (STCNet), one of the well-known techniques for creating sign language videos, enhances the accuracy and consistency of generated finger gestures by preserving temporal consistency and optimizing body shapes. This network has enhanced video-based sign language production (Jointly Harnessing Prior Structures and Temporal Consistency for Sign Language Video Generation) (Suo et al. 2024) by showcasing its efficacy on many datasets, such as WLASL-2000 and LSA64. Diffusion-based models, such as the Neural Sign Actors system, have evolved to further improve realism in sign language creation, particularly in 3D avatar generation. These models create incredibly Lifelike 3D signing avatars from text inputs by using graph neural networks that are inspired by anatomy. (Neural Sign Actors: A Diffusion Model for 3D Sign Language Production from Text) (Baltatzis et al. 2024) sets a new standard for Sign Language Production by improving semantic correctness and visual realism through the inclusion of 4D signing data. When taken as a whole, these efforts provide a direct path towards increasing the accessibility and accuracy of sign language generating systems. By adding dynamic 3D animation production, your suggested system expands on previous breakthroughs by converting Bangla text and speech into 3D Bangla Sign Language utilizing HamNoSys to SiGML conversion. With a focus on BdSL, this strategy can close current gaps in inclusion and accuracy.

Numerous research has examined the particular difficulties that arise when cross-modal machine translation (MT) is used in spoken and sign language translation systems. To solve the challenges of simultaneous morphology in sign languages, SR Bhagwat et al. translate Marathi into Indian Sign Language (ISL) (Bhagwat et al. 2023). Sign languages employ manual, non-manual, and spatial characteristics concurrently, in contrast to linear spoken languages. The study highlights the necessity of frameworks that support these complex emotions to improve communication across communities of people with hearing impairments. This is consistent with our work on BdSL, where we aim to bridge modality-specific gaps through smooth translation utilizing HamNoSys to SiGML conversion. By utilizing hierarchical memorized context in dialogue-based situations, L Gao et al.'s SLQA system improves sign language translation (SLT) (Gao et al. 2024). The architecture incorporates pairs of multimodal question-answering (QA) to guarantee logical consistency in conversations. This is consistent with our system's focus on the precise and continuous creation of

BdSL gestures, guaranteeing language coherence in animated scenes. By using contextual data to increase translation accuracy, these systems improve user engagement by producing useful results. By integrating a cross-attention mechanism and auxiliary tasks like spatio-temporal semantic modeling (SSM) and masked language modeling (MLM), K Liu et al. emphasize visual context learning in continuous sign language recognition (CSLR) (Liu et al. 2024). These activities fill up the semantic void between natural language text and video glosses. In a similar vein, our technology combines visual and linguistic information to produce accurate BdSL gestures, improving the efficiency and usability of communication for deaf groups that speak Bangla. Collectively, this research highlights the significance of cross-modal knowledge, simultaneous expression processing, and context-aware frameworks. By using such advancements, our BdSL system bridges the gaps created by conventional techniques by offering Bangla-speaking deaf users an engaging, effective, and easily accessible tool.

The development of deep learning models, optimization techniques, and machine learning has had a major impact on gesture detection systems. We want to improve the 3D gesture animations' precision, consistency, and quality by implementing these cutting-edge methods. Recent developments in deep learning have had a big influence on a lot of different sectors, such as urban planning and medical picture analysis. For example, researchers in Nature Medicine (2024) suggested a deep learning system for predicting the advancement of diabetic retinopathy, showing great accuracy in early detection and progression prediction (Dai et al. 2024). Similarly, UrbanEvolver has investigated urban layout regeneration using function-aware techniques, demonstrating the promise of deep learning in optimizing urban settings (Qin et al. 2024). These works lay the groundwork for the broad use of deep learning in various fields, which inspired our efforts to create 3D gesture animations in Bangla Sign Language through the use of an innovative conversion method from HamNoSys to SiGML. Also, some current methods focus on increasing scalability, accuracy, and efficiency, particularly in sign language translation and human-computer interface. A hybrid system that combines deep neural networks (DNNs) with metaheuristic optimization was presented by Mohyuddin et al. in 2023 (Mohyuddin et al. 2023). It selects features from Leap Motion Controller data using the Spotted Hyena-based Chimp Optimisation Algorithm (SSC). This method outperformed methods such as Particle Swarm Optimisation and Grey Wolf Optimiser, achieving an astounding 98% accuracy. In a similar vein, Dang et al. (2022) created a two-pipeline architecture keypoint-based gesture recognition system that achieved up to 98% accuracy with little computing cost. Using SURF and BOVW models for sign language motions, SVM and CNN for classification, and real-time recognition, Katoch et al. (2022) concentrated on Indian Sign Language recognition. All of these approaches, which prioritize lightweight models, real-time processing, and feature selection, provide insightful information that may be utilized for systems such as animation in Bangla Sign Language. Significant improvements in accessibility for the deaf and speech-impaired communities are anticipated with the use of such hybrid techniques.

In this part, we intend to present a complete review of existing research using the Hamburg Notation System (HamNoSys) and Signing Gesture Markup Language (SiGML), as well as other relevant methodologies in sign language production and 3D avatar construction. Siegmund Prillwitz created the HamNoSys notation in 1984 (Hanke 2004). A popular notation system for sign languages is called HamNoSys. It is a common transcription of around 200 characters that may be used to symbolize SL anywhere (Sugandhi and Kaur

2020). It offers a standardized approach to describing the gestures used in sign language, including hand movements, facial expressions, and hand shapes. Koller et al. (2016) discussed current improvements in continuous sign language recognition, emphasizing the possible integration of articulator-level subunit classifiers, similar to HamNoSys. Despite its value, issues remain in properly labeling real-world circumstances, highlighting the need for future development in language-agnostic systems like HamNoSys, as stated by many researchers (Hanke 2004; Koller et al. 2016; Kumar and Kaur 2018). Arkushin et al. pioneered the animation of text into signed postures using HamNoSys, offering a world-wide technique independent of destination sign languages (Arkushin et al. 2023). Elliott et al. (2004) created SiGML, a specialized markup language that helps specify sign language sequences for computer-generated performances. A standardized markup language called SiGML was developed to represent sign language gestures in digital form. It offers a methodical technique to describe hand gestures, facial emotions, and other aspects of sign language. SiGML is used to annotate or mark up sign language movements in a machine-readable format and is based on XML (eXtensible Markup Language). It begins with the tag and concludes with the closing tag. It is a flexible tool for sign language technology developers, researchers, and educators since it supports the depiction of a variety of sign languages. SiGML has been used in a variety of applications, including translating utterances into sign languages (Sanaullah et al. 2022; Aliwy et al. 2021; Sugandhi and Kaur 2020; Badhe and Kulkarni 2015) and improving animation realism and efficiency (McDonald et al. 2016). Sanaullah et al. (2022) proposed an architecture for translating utterances into Pakistani Sign Language (PSL), whereas Aliwy et al. (2021) used SiGML and HamNoSys to develop a dictionary converting Arabic text into Arabic Sign Language (ArSL). Sugandhi and Kaur (2020) presented a system for translating English text into Indian Sign Language (ISL) using SiGML, allowing human-computer interaction without an ISL translator. Throughout the literature, SiGML emerges as a standardized markup language critical for encoding animations in sign language, allowing for the smooth integration of sign language into diverse technological applications. Notable contributions are the translation of ISL signals into English (Badhe and Kulkarni 2015) and approaches for improving animation realism without manual detailing (McDonald et al. 2016). However, while previous research gives useful insights, more extensive analysis is required to uncover existing gaps and weaknesses. By highlighting these flaws, we hope to emphasize the importance and originality of our work in advancing the fields of sign language production and 3D avatar development.

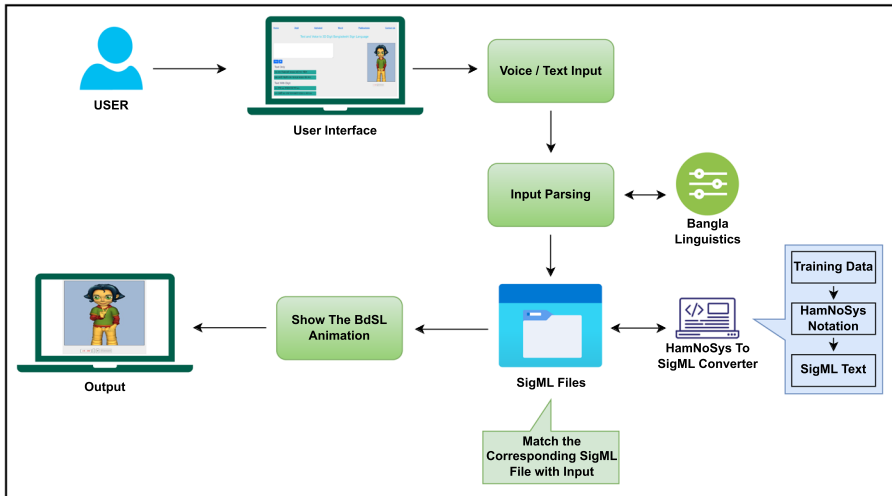
Recent improvements in sign language technologies are aimed at increasing accessibility for the Deaf and Hard-of-Hearing (DHH) community. Two important contributions are SignAvatar (Dong et al. 2024) and IsharaKotha (Rahman et al. 2023). SignAvatar uses a transformer-based Conditional Variational Autoencoder (CVAE) to reconstruct and generate 3D sign language movements. It employs CLIP for semantic conditioning and a curricular learning technique to improve motion realism. When tested on the ASL3DWord dataset, it outperforms other methods in producing isolated sign words. IsharaKotha introduces the HamNoSys-based Bangla Sign Language (BdSL) corpus, which has around 3800 words. It creates an avatar-based system for real-time BdSL creation utilizing SiGML and JASigning, resulting in high-quality animations with an average rating of 3.32/4.00 in expert reviews. SignAvatar specializes in 3D motion synthesis, whereas IsharaKotha offers a basic corpus for BDSL. Together, these activities help to achieve the larger objective of improving communication accessibility for the DHH community through creative technical solutions.

In conclusion, signs have traditionally served as a way of communication in the lack of standardized spoken languages, as proven by their usage by Middle Ages monks who took vows of silence. In 1620, Spanish priest Juan Pablo designed a curriculum for sign language users. In 1771, Abbe Charles established the first deaf school in Paris, standardizing sign language. Daniels (1997). This Literature review digs into the developing topic of automated sign language development, highlighting the scarcity of research on 3D animated sign language motions generated from text and spoken inputs. To close this gap, we need to investigate transcription standards like HamNoSys and use SiGML for sign language animations (Hanke 2004; University of Hamburg. Sigml 2025). Ferlin et al. (2023) emphasize ongoing challenges, particularly with inaccurate labelling for practical applications. The Literature review emphasizes the historical history of sign language communication, the lack of research on 3D animated sign language motions, and the relevance of investigating transcription standards such as HamNoSys and SiGML. The report highlights the continued development of sign language technology and ongoing initiatives to improve inclusion and communication.

### 3 Proposed model for 3D animated gesture generation in BdSL

Our proposed system's architecture and parts are outlined in a research framework that we proposed, building on the knowledge from the literature review. The conversion of HamNoSys to SiGML is a critical part of our research. We use the JA SiGML URL App as our avatar that makes animations. We give an overview of the HamNoSys notation and how it functions to describe gestures used in sign language. First of all, the suggested system takes input from the user as text or voice. Then the system parses the input using Bangla Linguistic rules. Root words with the corresponding morphological information are produced after parsing. BdSL rules are applied for each input using this information. The word order is retrieved, and an output is produced using these BdSL Linguistics principles. A separate system has already been created to generate SiGML from HamNoSys, and this is open to all. The appropriate HamNoSys for every root word is generated beforehand, transformed into SiGML tags, and saved as a SiGML file. Then, the SiGML files were saved in the dataset folder. Every time a word is requested, the corresponding SiGML file is then sent to the animation server. The SiGML file's tags determine how animation frames are created and combined to animate the 3D avatar. Therefore, to communicate with the Deaf people, the suggested technology can play the signs automatically. Figure 2 depicts the architecture of the proposed system.

Figure 3 shows the flow of our proposed system. The system flow diagram demonstrates various modules, each of which has unique interactions and functions. The system parses the user's input, which can be a sentence, a word, an alphabet, or a number, which can be in the form of only text, text with digits, or only digits, before going to further processing. After parsing the input, the suggested system removes all space, punctuation marks, etc., and divides the input into corresponding tokens (Root Word). Then, the token searches the corresponding SiGML file by Asynchronous API (a programming paradigm that allows multiple operations to be performed concurrently without blocking the execution flow) in the dataset folder. After that, if the tokens (like Word or Digit) match any SiGML file in the dataset folder, then this file is sent to the sequence generator; if they don't match, then the



**Fig. 2** System architecture of the proposed system for sign language animation generation from voice or text

system again sends it to the parsing process and converts it into the alphabet. Then again, the system sends the SiGML file for Alphabet to the sequence generator so that the system can spell the input. After that, the process of sequence generation and sign production starts. Algorithm 1 shows the process of finding the corresponding token for BdSL Animation. If the system doesn't find the word in the dataset, then it converts the words into the alphabet and produces the sign by spelling the word. A file in SiGML format is small enough to send over the internet and simple enough to render using a 3D avatar on a website. The process can be called the Bangla Word Spelling System in BdSL.

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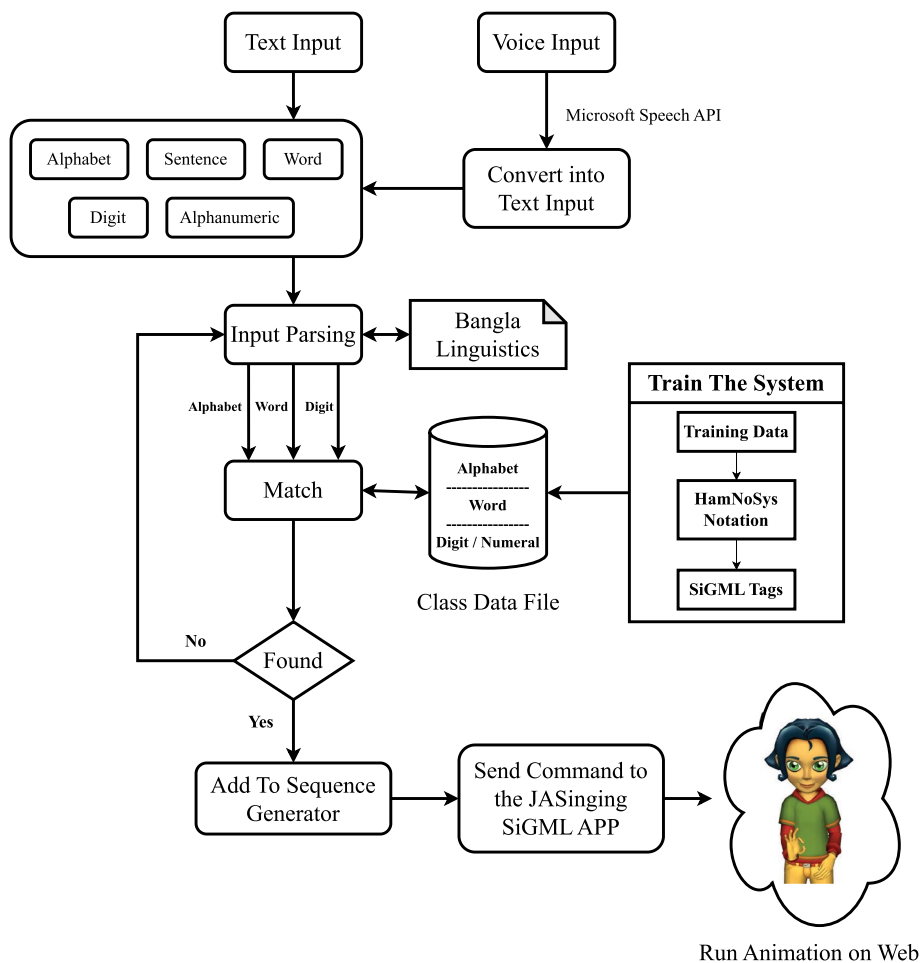
1: Input: Parsing Sequence[]
2: Output: Playing Sequence[]
3: Initialization: Input Sequence [i] as Sequence, Initial Variable  $\theta$ , Playing Sequence  $\pi$ , Alphabet Dataset  $\chi$ 
4: function PLAYSEQUENCE():
5:   for Every Token in Sequence do
6:     Choose Relevant BdSL;
7:     Generate Tokens;
8:     if Token == BdSL then
9:        $\pi :=$  Token;
10:    else
11:       $\pi := \chi[\text{Token}]$ ;
12:    end if
13:  end for
14:  return Playing Sequence for BdSL;
15: end function
  
```

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**Algorithm 1** Finding tokens for BdSL animation generation

### 3.1 Input types and processing

Our proposed system receives input from two environments: (i) speech and (ii) text. The Microsoft Speech Recognition API is used to convert voice into text input. The suggested system requires five different kinds of inputs: Bangla Sentences, Bangla Words, Bangla



**Fig. 3** Process flow diagram of the proposed system for sign language animation generation from voice or text

Alphabet, Bangla Digits, and Bangla Composite Numbers. There are also three types of composite numbers. These are Text with Digit Input, Text Only, and Digit Only. The system delivers the input to the Parsing procedure after receiving the input (Sikder et al. 2024; Das et al. 2024). Our system's input types and input examples are shown in Table 2.

### 3.2 Parsing process for inputs in bangla text of the proposed system

Firstly, the suggested system eliminates punctuation and white spaces throughout the parsing process. Next, inputs are transformed into root words. The parsing process uses Bangla Linguistic rules to complete the process. The root words are ০ (Zero) to ৯ (Nine), হাজার (Thousand), লক্ষ (Lakh), কোটি (Crore), Bangla Alphabets, and the most commonly used 40 Bangla words. The technique that is suggested divides the text into tokens that represent a single text. For example, the numeral আটচল্লিশ কোটি ছাব্বিশ লক্ষ পয়ষট্টি হাজার চার

**Table 2** Input types and corresponding examples in the proposed system dataset

| Input Types                                | Examples Inputs in Bangla                                                                                                                    |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Bangla Sentence                            | আমার নাম আরফি (My name is Arif)                                                                                                              |
| Bangla Words                               | আমি (I), নাম (Name), খাওয়া (Eat)                                                                                                            |
| Bangla Alphabet                            | ক (Ka), থ (Th), ন (N), ম (M), আ (A), উ (U)                                                                                                   |
| Bangla Digit                               | ১ (1), ৭ (7), ৫ (5), ৩ (3), ৮ (8)                                                                                                            |
| Bangla Composite Numbers (Only Text)       | আটচল্লিশ কেরাটি ছাব্বিশ লক্ষ পয়ষট্টি<br>হাজার চার শত ছত্রিশ (Forty-Eight Crore Twenty-Six Lakh Sixty-Five Thousand Four Hundred Thirty-Six) |
| Bangla Composite Numbers (Text with Digit) | ৪৮ কেরাটি ২৬ লক্ষ ৬৫ হাজার ৪৩৬ (48 Crore 26 Lakh 65 Thousand 436)                                                                            |
| Bangla Composite Numbers (Only digit)      | ৪৮২৬৬৫৪৩৬ (482665436)                                                                                                                        |

শত ছত্রিশ (Forty-Eight Crore Twenty-Six Lakh Sixty-Five Thousand Four Hundred Thirty-Six) is broken into tokens চার (Four), আট (Eight), কেরাটি (Crore), দুই (Two), ছয় (Six) লক্ষ (Lakh), ছয় (Six), পাঁচ (Five), পাঁচ (Thousand), চার (Four), তিন (Three), ছয় (Six) by the suggested system. Then arrange them into the right order using Linguistic Rules so that the animation plays in the right order.

Figure 4 shows the Bangla input parsing. The word parsing occurs if the system doesn't find the word in the database. The Proposed system applies two different types of algorithms to complete the overall parsing process. In Algorithm 2, we show that the system predicts the voice input and converts it into text. Then the parsing process starts. The system parses the inputs according to the Bangla Linguistics. The system converts the composite numerals into individual tokens using the Predict(); function for voice and text input. Algorithm 2 works for only Bangla Numeral parsing.

```

1: Input:  $\alpha\beta[]$ 
2: Output: Display Animated(3D) Sign in Bangla for Bangla Numeral []
3: Initialize: Voice ( $\alpha$ ), Text ( $\beta$ ), Recognition Speech ( $\gamma$ ), Parsing Input [ $i$ ], Primary Size ( $n$ ), Primary Location 0 to  $n$ , Primary Variable ( $a, b$ )
4: function PREDICT  $\alpha(i)$ :
5:   for  $i$  from 0 to  $n$  do
6:     if  $\gamma[i]$  equals  $\beta$  then
7:       ApplyAlgorithm[];
8:       Return [True];
9:     end if
10:    Return [False]
11:  end for
12: end function
13: function PREDICT  $\beta(i)$ :
14:  for  $i$  from 0 to  $n$  do
15:    if  $a$  equals  $n/i$  then
16:       $b = a*i$ ;
17:      Return [ $b$ ];
18:    end if
19:  end for
20:  for  $i$  from 0 to  $n$  do
21:    if  $a$  equals  $n/i + 1$  then
22:       $b = a*(i + 1)$ ;
23:      Return [ $b + 1$ ];
24:    end if
25:  end for
26: end function

```

**Algorithm 2** Bangla composite numerical parsing

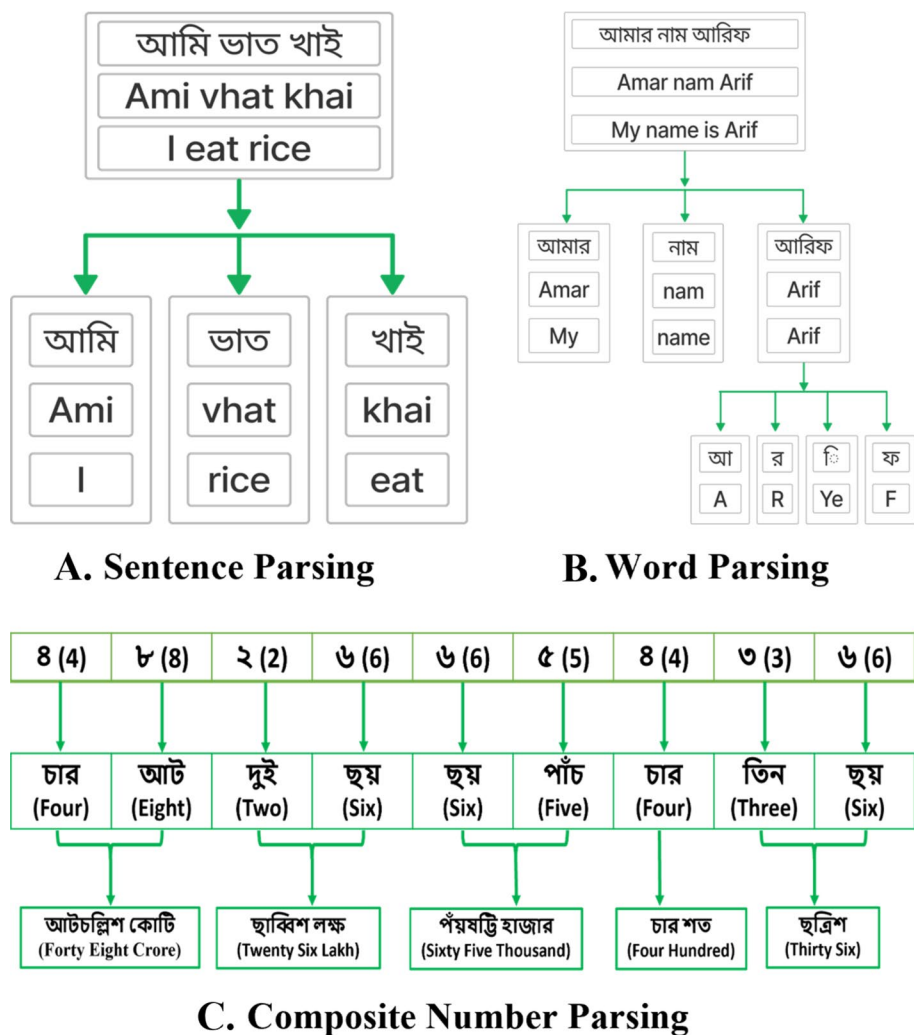


Fig. 4 Example of input parsing process for the proposed system

Algorithm 3 is used for sentence and word parsing. The proposed system divided the sentence into words by the space between the words in the sentence. Then the system searches for the word token to generate a sign. If the system finds the word token, then it runs the animation. Otherwise, the system is parsing the word also into the alphabet. And then generate alphabet tokens and spell the word. The system parses the sentence by using the SentenceParsing(); function and parses the word by using the WordParsing(); function. The SentenceParsing(); function converts the sentence into a word, which is a token, and the WordParsing(); function converts the word into the alphabet, which is also a token. And then Algorithm 1 is work. By this process, the system can run complex words and show the animation by spelling the complex word.

---

```

1: Input:  $\alpha\beta[]$ 
2: Output: Display Bangla Animated(3D) Sign for Bangla Sentence and Word []
3: Initialize: Voice  $\alpha$ , Text  $\beta$ , SentenceParsing  $\delta$ , WordParsing  $\sigma$ 
4: function SENTENCEPARSING $\delta$ (Text):
5:   for Each Word in Text do
6:      $\omega^+ = \text{word}$ ;
7:     return  $\omega$ ;
8:   end for
9: end function
10: function WORDPARSING $\sigma$ (Text):
11:   for Each Character in Text do
12:      $\omega^+ = \text{character}$ ;
13:     return  $\omega$ ;
14:   end for
15: end function

```

---

**Algorithm 3** Bangla sentence and word parsing

### 3.3 HamNoSys notations to represent sign language gestures

The Hamburg Notation System (HamNoSys) (Hanke 2004), delivers a standardized method for characterizing the hand forms, orientation, placement, and movement as well as non-manual elements (including facial emotions) that are utilized in sign language. These notations can be used to automatically create animated gestures by providing a formal representation of sign language gestures.

Figure 5 demonstrates the fundamental hand shapes found in HamNoSys notation from Hanke (2004). These forms are essential for explaining sign language motions since they depict the hand's structure (open hand, fist, extended index finger, etc.).

Figure 6 illustrates the HamNoSys notation's hand orientation (Hanke 2004). For motions to be appropriately represented, this includes the orientation in which the palm or fingers are oriented (palm up, palm down, fingers to the side, etc.).

Figure 7 shows how to move the hands in HamNoSys notation (Hanke 2004). To convey the dynamic elements of sign language, movements—which might be linear, circular, or complicated trajectories—are crucial.

Figure 8 outlines the overall framework of HamNoSys notations (Hanke 2004), which integrate non-manual elements, hand forms, hand orientation, hand location, and hand movement into a single representation. Researchers can formalize sign language motions using this structure so that computer systems can process them.

We make sure the movements are expressed in a uniform and machine-readable way by developing HamNoSys notations for our training data, which is an essential step in the automatic creation of sign language animations.

### 3.4 SiGML language for sign language animation generation

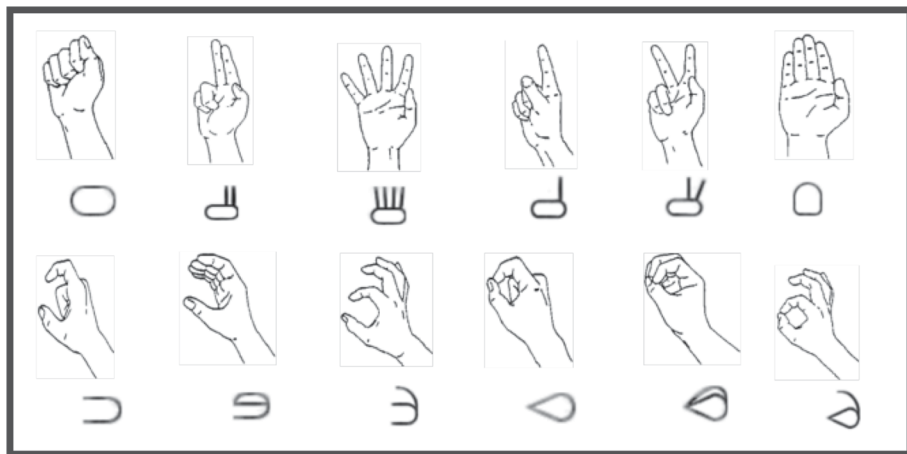
Once the HamNoSys notations for the training data have been created, they must be converted into SiGML (Signing Gesture Markup Language), a markup language created especially for expressing sign language gestures in a way that 3D avatar software can understand. An illustration of SiGML tags for the Bangla digit "दूहै (Two)" is shown in Fig. 9. In a structured XML-like style, the SiGML tags encode the hand shape, orientation, location, and movement of the gesture, together with non-manual elements. This enables the motion to be faithfully rendered as an animated sign by the 3D avatar program.

A crucial phase in our system is the conversion from HamNoSys to SiGML, which fills the gap between the formal representation of gestures (HamNoSys) and their actualization as dynamic animations (SiGML). Natural and fluid sign language animations are produced by using the SiGML format, which guarantees that the movements are shown with the appropriate timing, spatial relationships, and non-manual aspects.

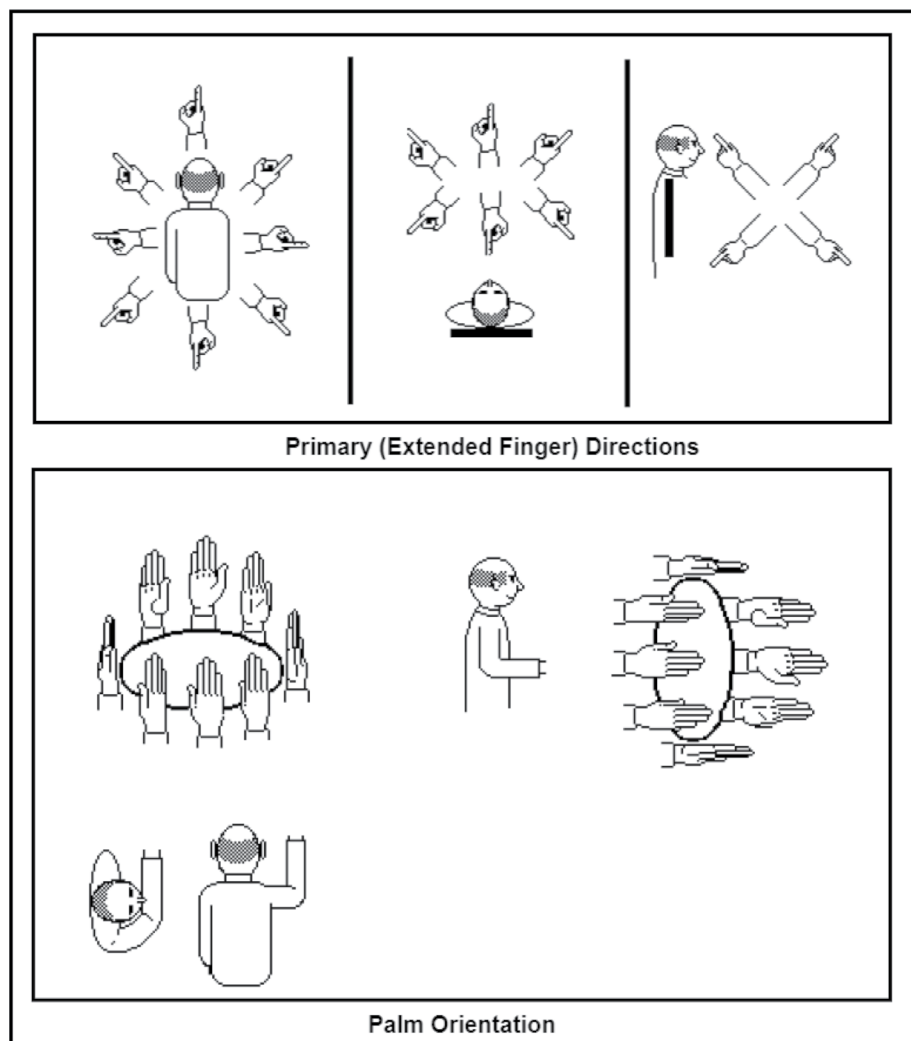
### 3.5 Sequence generator

It is a crucial module for the system that is being suggested. When we find the corresponding SiGML file, it needs to be sent to the animation server sequentially to run. Therefore, when tokens are converted to their basic form, this module arranges every word sequence to play BdSL signs in accordance following sequence. Bangla sign language grammar rewriting rules are used to guarantee that BdSL words, phrases, and numbers have the correct structure. A BdSL phrase, word, number, or structure can be formed from any plausible combination of signs.

In summary, the SiGML representation produced from the HamNoSys sign language notation can be read by the 3D avatar software. Algorithm 1 shows the process of finding the corresponding token for BdSL Animation. The conversion process from HamNoSys to SiGML is a crucial part of the technology used in the research. The simple conversion steps of BdSL to animation are shown in Fig. 10.

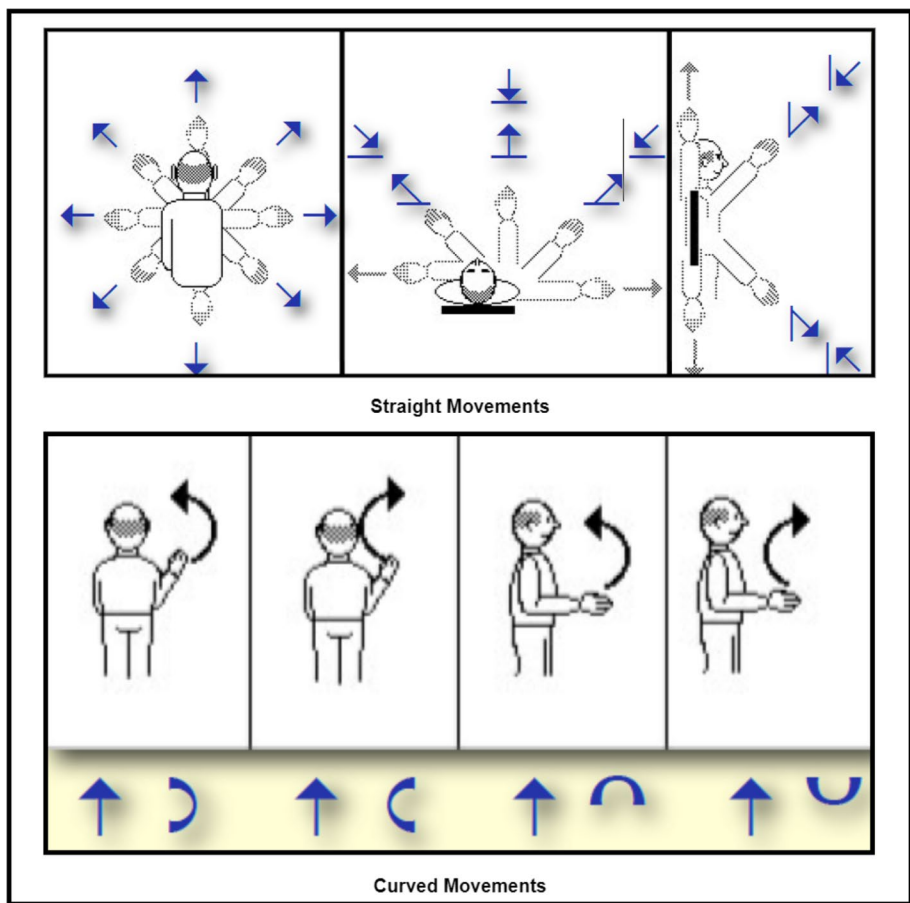


**Fig. 5** Basic hand shapes of HmaNoSys notations to represent sign language gestures

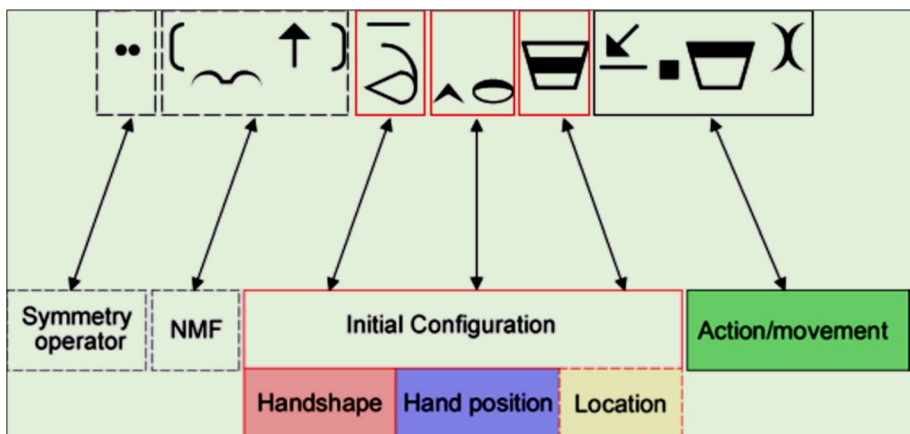


**Fig. 6** Hand orientation of HmaNoSys notations to represent sign language gestures

Overall, through a logical strategy, the research developed a technique and suggested a procedure for developing an automated system that can create animated (3D) BdSL gestures from Bangla voice and text input. The methodology described highlights the potential for real-world applications in the areas of assistive technology, communication accessibility, and education for those with hearing impairments or those interested in learning the Bangla Sign Language. It also shows a deep understanding of both sign language linguistics and technological implementation.



**Fig. 7** Hand movement of HmaNoSys notations to represent sign language gestures



**Fig. 8** General structure of HmaNoSys notations to represent sign language gestures

**Fig. 9** Example of SiGML tags for Bangla Digit "দুই (Two)"

```
<sigml>
<hns_sign gloss="two">
  <hamnosys_nonmanual>
  </hamnosys_nonmanual>
  <hamnosys_manual>
    <hamfinger23spread/>
    <hamthumbacrossmod/>
    <hamextfingeru/>
    <hampalmd/>
    <hamshoulders/>
    <hamclose/>
    <hammoveo/>
    <hamsmallmod/>
  </hamnosys_manual>
</hns_sign>
</sigml>
```



**Fig. 10** Sequence for BdSL animation generation of the proposed system

## 4 Results analysis and discussion

A critical component in determining the effectiveness, accuracy, and usability of the established system in the research on "Bangla Sign Language 3D Gesture Animation Generation Using Hamnosys to SiGML Conversion" is the evaluation of the system. The system's performance is evaluated, and its advantages are analyzed in this part. Output analysis, assessment metrics, comparison with other systems, and user studies are often some of the major parts of the evaluation process. To guarantee that the evaluation appropriately depicts the system's performance across multiple input conditions, it is crucial to choose the right sign. The system's performance can be objectively evaluated with the help of assessment metrics. These metrics might include indicators of correctness, including the proportion of correctly generated gestures in comparison to real-world or human annotations.

According to our findings, gesture animations have significantly improved in terms of temporal consistency, which is comparable to developments in video processing methods. As an illustration of the potential of wide learning frameworks in preserving continuity and minimizing artefacts, a broad learning system has been used to enhance video temporal consistency (Sheng et al. 2021). Using similar concepts, our method guarantees realistic and fluid gesture animations. Additionally, measures like resource consumption, processing speed, and computational efficiency may be considered. User studies are crucial to assess the system's usability and user experience. This may entail obtaining input from BdSL

expert who engages with the system and offers their subjective opinions of how well it functions. The constraints and potential sources of error of the proposed system should also be evaluated. To emphasize the uniqueness and improvements of the established system, comparisons with other methods or systems for producing sign language gestures can also be made. Benchmarking against other systems or conducting comparative studies may be used in these comparisons to show the superiority or distinctive contributions of the suggested strategy.

## 4.1 Experimental setup

We use a Plextone G30 PC Gaming Headset with a built-in microphone in our evaluation setup because of its excellent speech-recording capabilities. This high-performance audio equipment is essential to our system's ability to collect speech inputs clearly and accurately. We utilize the processing capability of an Asus VivoBook S15 S533EQ, which is outfitted with a Core i5 11th Gen CPU and an MX350 2GB graphics card, to process and run our system. The computational demands of our suggested system are satisfied by the reliable and efficient performance provided by this cutting-edge hardware arrangement. We chose Visual Studio Code, a popular and adaptable integrated development environment (IDE), to develop the system. Importantly, accessibility and user-friendliness are given top priority in our system design. The UEA (University of East Anglia) avatar animation engine is smoothly interfaced with this graphical user interface (GUI), which is intended to automate the animation production process.

## 4.2 Datasets description

We used 90 data classes to prepare the system, including 41 Bangla word data classes, 36 Bangla letter data classes, and 13 Bangla numerical data classes. SiGML tags are used to express the distinct data that each class has, including Hand Shape, Hand Orientation, Hand Movement, and Notations. We looked through existing English HamNoSys datasets (about 6,000 data points) for Bangla alphabets, words, and numbers that have HamNoSys notations. Nevertheless, it was discovered that only 20% of the data resembled Bangla Sign Language (BdSL). We developed the remaining 80% of the HamNoSys notations, especially for BdSL, and we adjusted the corresponding 20% to fit BdSL gestures better. The dataset classes were then created by converting these notations into SiGML. Zenodo provides the SiGML datasets used in this investigation at <https://doi.org/10.5281/zenodo.12152444>. This guarantees openness and enables other scholars to confirm and expand on our findings. Our system's total number of data classes and the number of input uses for system testing are shown in Table 3.

The system can produce animations for every numeric value according to the 13 trained Bangla numeric data classes displayed in Fig. 11.

The system can spell every Bangla word or sentence thanks to the 36 learned Bangla alphabet data classes shown in Fig. 12.

Ten instances of trained Bangla word data are shown in Fig. 13, showcasing the system's proficiency with intricate word movements.

We guarantee that the system is inclusive and reliable for a broad spectrum of users by taking into account user diversity and addressing any biases during dataset creation and testing.

### 4.3 Performance evaluation parameter

A precise assessment is essential to comprehending the advantages and disadvantages of our suggested strategy. The chosen performance assessment measure is a quantitative indicator that provides information about the accuracy, consistency, and general efficacy of our system for generating BdSL animations. Equation 1 is used to compute the accuracy. Where  $q$  is the total number of inputs and  $p$  is the number of animations that are appropriately created for a particular input. We evaluated the generated sign by a sign language expert to evaluate the appropriateness of the generated sign. Additionally, we have contrasted the generated sign with the dictionary of the Bangla sign language. Then, we count the value of  $P$ . In this case,  $q = 100$ .

$$Accuracy = \frac{p}{q} \times 100\% \quad (1)$$

### 4.4 System's performance

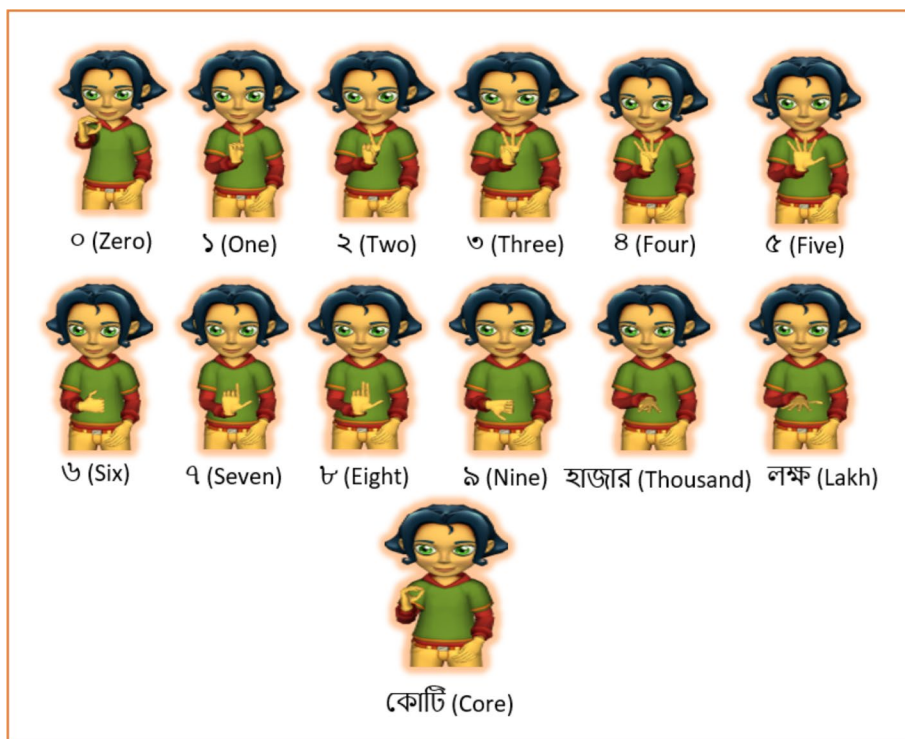
The system has been tested 500 times for Bangla text input and 10 times for each voice input of numbers, letters, and words. Table 4 presents the system's average accuracy for tests of each Bangla composite numeral, sentence, word, and alphabet as text input and voice input. When Bangla words, alphabets, and digits are entered into the system, it remains unfounded. However, it occasionally generates inaccurate sign animations for several Bangla compound letters and complex words. Even though we have a spelling system in place, the system has trouble correctly identifying some compound characters; as a result, the average accuracy for all forms of Bangla text input is 97.50%. The system's accuracy for voice input is a little lower at 94.75%, mostly because of issues with background noise, accent changes, and dialect diversity.

We carried out system testing to present a more thorough examination:

- User-based testing: To guarantee inclusivity and robustness, we performed user-based testing with 90 native Bangla speakers from various demographic and geographical backgrounds, including Dhaka, Chittagong, and Sylhet (30 people from each district). Because Bangla is a complicated language with large regional differences in accent and dialect, we gathered voice samples from speakers of the three main dialect groups in Bangladesh: Dhaka (standard Bangla), Chittagong, and

**Table 3** Total number of data classes and number of input uses for the proposed system

| Data Class Type | Number of Data Classes | Number of Text Input for Testing | Number of Voice Input for Testing |
|-----------------|------------------------|----------------------------------|-----------------------------------|
| Bangla Sentence | 0                      | 150                              | 300                               |
| Bangla Words    | 41                     | 230                              | 370                               |
| Bangla Alphabet | 36                     | 45                               | 76                                |
| Bangla Numeral  | 13                     | 95                               | 155                               |



**Fig. 11** 13 Trained data for bangla numeric

Sylhet. Also, we divided each group of people into 3 age groups. These are the 15-25, 26-40, and 41-55 age groups. There is a mix of men and women in each group. Each person provided an equal number of samples (10 voice inputs) to guarantee a balanced representation. The results showed that the system worked best with the Dhaka dialect (standard Bangla), with an accuracy of 96.50%; for the Chittagong and Sylhet dialects, the accuracy decreased to 93.50% and 92.00%, respectively, due to differences in pronunciation and intonation patterns. This method was useful in capturing changes in accent, dialect, and signing approaches. During testing, people engaged with the system by entering text and voice and assessing the resulting animations. The testing dataset is available on the sheet: <https://docs.google.com/spreadsheets/d/1y5BPlc6uQWV5-Odx9Lq6XdJPn9ad2pjM34yxYJCvBsY/edit?usp=sharing> Noise impact: We evaluated the system in a range of background noise levels, from quiet to moderately noisy (e.g., street noise, office noise) locations. According to the findings, the accuracy of the system decreased by approximately 3–5% in loud circumstances as opposed to quiet ones. This is because background noise may cause the voice recognition module to misinterpret input.

- Resolving potential biases: The suggested technique automatically reduces biases relating to geographical variances, dialects, or individual signing styles because it is based on standard Bangla Sign Language (BdSL) gestures. The adoption of a



Fig. 12 36 Trained data for bangla alphabet



Fig. 13 Example of 10 trained data for bangla word

**Table 4** The accuracy of the suggested system for generating BdSL animations

| Input Type                                | Text Accuracy (%) | Voice Accuracy (%) |
|-------------------------------------------|-------------------|--------------------|
| Bangla Composite Number (Only Digit)      | 100               | 96                 |
| Bangla Composite Number (Digit with Text) | 100               | 96                 |
| Bangla Composite Number (Only Text)       | 95                | 96                 |
| Bangla Sentence                           | 94                | 91                 |
| Bangla Word                               | 96                | 94                 |
| Bangla Alphabet                           | 100               | 98                 |
| Average                                   | 97.50             | 94.75              |

defined set of gestures guarantees that the system is not biased toward any one dialect or signing style, even though BdSL itself may have certain regional variances.

- **BdSL expert testing:** To assess the accuracy and efficacy of our system, we conducted a thorough test with a qualified Bangladeshi Sign Language (BdSL) expert. The system received a total of seven forms of input (given in Table 2), and we gave ten inputs for each category of input, resulting in sign language outputs. The BdSL expert meticulously analyzed each output gesture, evaluating its syntax, clarity, and conformance to normal BdSL rules. The system correctly processed all inputs, and the BdSL expert commented "Pass" on each one, validating the system's capacity to accurately convert Bangla text into BdSL. Also, he gives some feedback about the system. Feedback from BdSL experts indicated that the algorithm properly created standard gestures for the majority of input types. Users praised the 3D animation capabilities but also pointed out areas for development, notably the lack of non-manual signals (such as facial emotions), which would increase the animations' naturalness. In addition, several difficulties were encountered in producing complicated words and compound letters. To reduce biases and create a more inclusive system, future iterations will include community-driven validation, in which BdSL users and sign language specialists regularly analyze and revise gesture mappings. This iterative feedback process will help resolve any biases caused by geographical variances, dialects, or individual signing styles, resulting in a larger and more representative sign language dataset.

Considering these difficulties, the system performs admirably, achieving an overall voice input accuracy of 94.75%. Future research will enhance the system's resilience to noise and accent changes by utilizing dialect-specific training data and sophisticated noise reduction algorithms.

Figure 14 shows the proposed system interface. We included six pages here. The word page accepts word and phrase input, the digit page only accepts numeric input, and the alphabet page only accepts alphabet input. Users have the option to provide voice or text input. The user also adjusts the avatar's speed.

Some outputs from our system are shown in Figs. 15, 16, and 17. Figure 15 shows the output of Bangla Composite Number "৪৮ কোটি ২৬ লক্ষ ৬৫ হাজার ৪৩৬ (48 Crore 26 Lakh 65 Thousand 436)". Figure 16 shows the spelling output of the Bangla word after parsing from the Bangla name "তাসমিয়া (Tasmia)". And lastly, Fig. 17 shows the output of the Bangla Sentence "আমার বই এখানে (My book is here)".

Table 5 represents the computational cost of our system for the Bangla Composite Number and Bangla Sentence. For every type of input, we calculate the average computational cost by taking the input 10 times. Our suggested system is composed of many modules, as shown in the methodology. Consequently, the methodology is broken down into stages, each of which requires a certain amount of time to complete the animation. We calculated the computation cost by a JavaScript function called "performance.now()". The function automatically calculated the time for each single output. The web console shows the processing time when the system runs an input. The speed of the network determines this. During testing, the device utilized 2 MB of connections per second. As a result, values may change depending on the network speed. We find that the average computational cost for every input running our correct animation is 79.57 ms. A millisecond (ms) is a unit of time equal to one-thousandth of a second.

Figure 18 represents the speed of animation. How quickly the animation will be run is determined by the speed of the animation. The scale of speed is +3 to -3. The animation player defaults to setting the value to 0. We can control the speed manually. The animation runs at its fastest rate when the number is +3, and at its slowest pace when the value is -3. We take the output and test the system at a speed value of 0.

#### 4.5 Comparison analysis with state-of-the-art systems

For the deaf community of Bangla speakers, the study marks a breakthrough in accessibility and inclusion of communication. By introducing an innovative system that can automatically produce BdSL motions from text and speech input, our work addresses a significant technological gap and helps Deaf individuals communicate. Table 6 presents the findings of a thorough comparison study between the suggested system and state-of-the-art systems based on some important characteristics. The suggested system stands apart from current systems thanks to several special features. The capacity of the suggested system to analyze any kind of Bangla input, including inputs that are not in the dataset by spelling out the letters, is one of its main features. All of the systems under comparison lack this feature.

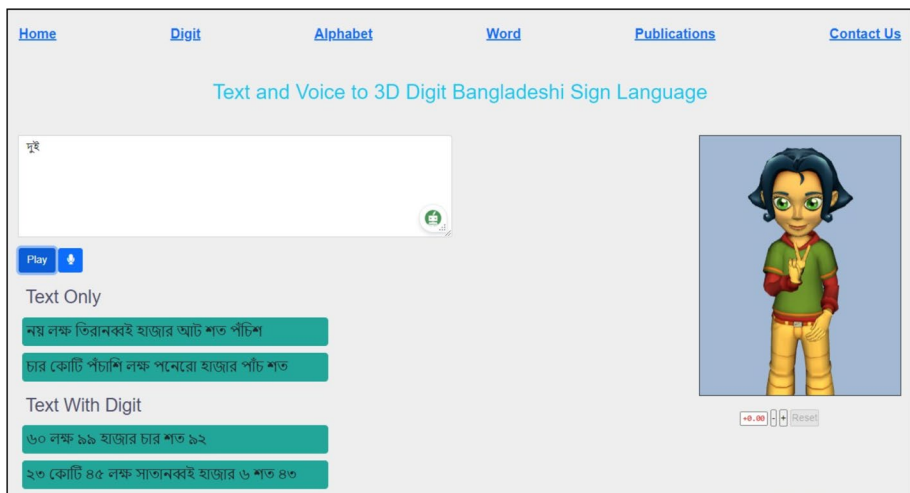


Fig. 14 Proposed system interface

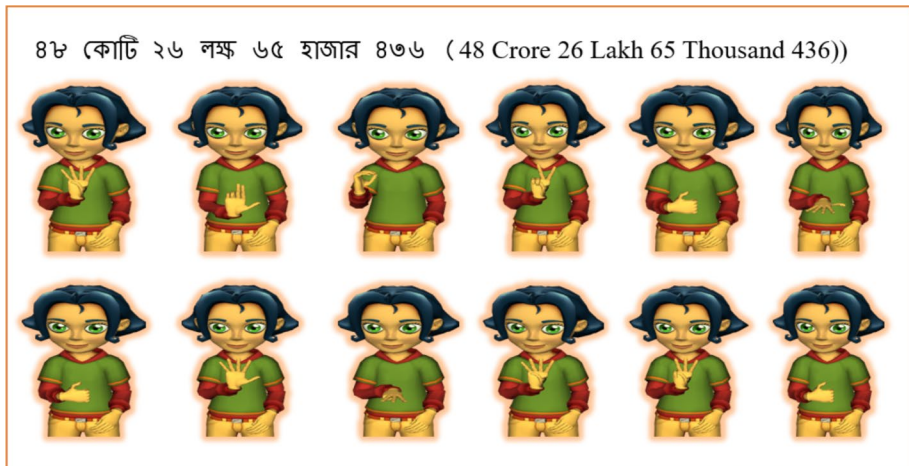


Fig. 15 Output animation for “৪৮ কোটি ২৬ লক্ষ ৬৫ হাজার ৪৩৬ (48 Crore 26 Lakh 65 Thousand 436)”

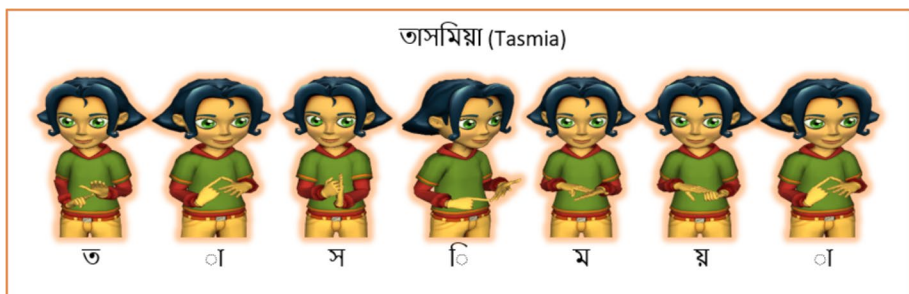


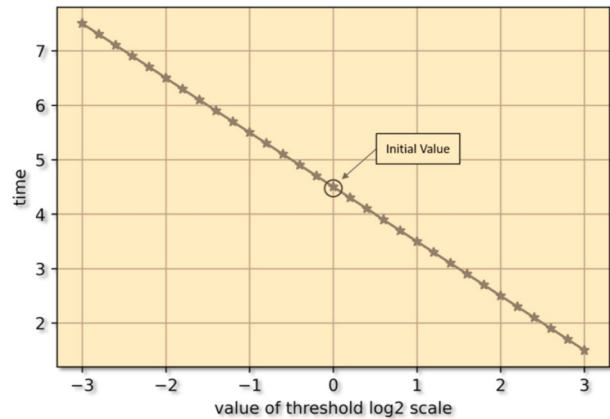
Fig. 16 Output animation for “তাসমিয়া (Tasmia)”

Fig. 17 Output animation for “আমার বই এখানে (My book is here)”



**Table 5** Computation costs for creating animated

| Input Type      | No. of Inputs | Average Computational Cost (ms) |
|-----------------|---------------|---------------------------------|
| Only Digit      | 10            | 42.10                           |
| Digit with Text | 10            | 267.65                          |
| Only Text       | 10            | 99.52                           |
| Bangla Alphabet | 10            | 9.45                            |
| Bangla Word     | 10            | 11.11                           |
| Bangla Sentence | 10            | 47.61                           |
| Average         |               | 79.57                           |

**Fig. 18** Animation speed graph plotting time against animation speed values

With a processing cost of only 79.57 ms per input, the system also achieves high accuracy rates of 97.50% for text input and 94.75% for voice input, making it extremely effective for real-time applications. As highlighted in Table 5, we present a quantitative comparison of important performance parameters, such as dataset size, input types, techniques and accuracy rates, to further emphasize the benefits of the suggested method. In terms of processing performance, input flexibility, and dataset diversity, the suggested system performs better than current solutions. Furthermore, the flexible API and spelling capabilities of the suggested system allow for future extensions, which are not present in any of the systems under comparison.

The suggested system not only enhances the dataset but also incorporates numerous novel features that considerably improve its performance and usability when compared to state-of-the-art methods. For example, the system's spelling capabilities enable it to handle any Bangla input, even if it is not explicitly included in the dataset, a feature that is missing from other comparable systems. Furthermore, the system's processing efficiency (79.57 ms per input) makes it ideal for real-time applications, beating systems such as Sanaullah et al. (2022) (88.3 ms for each phrase). Furthermore, the use of language parsing guarantees that the generated gestures are natural and fluent, addressing a major weakness in current systems that rely on direct text-to-gesture mappings. These enhancements collectively show the system's potential to overcome communication gaps more effectively than existing cutting-edge options.

**Table 6** Assessment of differences between the proposed work and existing system

| Term                | Proposed System                                        | Caballero-Morales and Trujillo-Romero (2013)    | Nawshin et al. (2020)            | Sugandhi and Kaur (2020)                    | Sanaullah et al. (2022)                                    | Aliwy et al. (2021) | Rahaman et al. (2020)               | Rahman et al. (2023)                          | Dong et al. (2024)                        |
|---------------------|--------------------------------------------------------|-------------------------------------------------|----------------------------------|---------------------------------------------|------------------------------------------------------------|---------------------|-------------------------------------|-----------------------------------------------|-------------------------------------------|
| 3D Animation        | Yes                                                    | Yes                                             | No                               | Yes                                         | Yes                                                        | Yes                 | No                                  | Yes                                           | Yes                                       |
| Voice Input         | Yes                                                    | Yes                                             | No                               | Yes                                         | No                                                         | No                  | Yes                                 | No                                            | No                                        |
| Text Input          | Yes                                                    | No                                              | Yes (Via OCR)                    | Yes                                         | Yes                                                        | Yes                 | Yes                                 | Yes                                           | Yes (CLIP Embeddings)                     |
| Bangla Input        | Yes                                                    | No                                              | Yes                              | Yes                                         | No                                                         | No                  | Yes                                 | Yes                                           | No                                        |
| Input Type          | All Kind of Bangla Input                               | Mexican Spoken Words and Sentences              | Scanned Images or PDF files      | English Words and Sentences                 | English Word, Numbers, Alphabet, and Sentences             | Arabic Word         | All Bangla Numeric                  | Bangla Text                                   | 2D Videos, Text, and Image                |
| Dataset Size        | Alphabet: 36, Word: 41, Numeral: 13                    | Word: 70, Sentence: 20                          | -                                | Word: 2,950, Sentence: 1000                 | Alphabet: 26, Numeral: 10, Word: 40, Sentence: 25          | 3D Sign: 3000       | Bangla Numeral: 14                  | BdSL Word: 3823                               | ASL Word: 103                             |
| Spelling Capability | Yes                                                    | No                                              | Yes                              | No                                          | No                                                         | No                  | No                                  | Yes                                           | No                                        |
| Technique           | HamNo-Sys, SiGML, BdSL Parsing, Avatar Based Animation | 3D Modelling Cap-ture, ASR, MLLR, HMM, HTK, BVH | Tesseract OCR                    | HamNo-Sys, SiGML, Rule-Based Parsing, WebGL | HamNo-Sys, SiGML, UEA Avatar, NLP                          | HamNo-Sys, SiGML    | Bangla linguistics for text parsing | HamNo-Sys, SiGML, RNN, Avatar-Based Animation | Transformer CVAE, CLIP, SMPL-X            |
| Accuracy (Text)     | 97.50%                                                 | -                                               | Character: 81.75%, Words: 61.87% | 0.95 BLEU                                   | Alphabet, Numeral, Words, and Phrase: 100%, Sentences: 80% | 86.00%              | 100%                                | 3.32 out of 4.00                              | Recons-truction: 97.6%, Generation: 73.3% |
| Accuracy (Voice)    | 94.75%                                                 | 96.20%                                          | -                                | -                                           | -                                                          | -                   | 96.03%                              | -                                             | -                                         |
| Processing Time     | 79.57 MS                                               | -                                               | -                                | -                                           | 88.3 MS                                                    | -                   | 9.5 MS                              | -                                             | -                                         |

## 4.6 Discussion

Our suggested method takes a huge step toward bridging the communication gap between deaf and hearing people by dynamically producing Bangla Sign Language (BdSL) animations from text and voice inputs. While the system is very accurate (97.50% for text and 94.75% for voice input), it is currently limited in its ability to handle complicated Bangla phrases, idiomatic expressions, and context-specific gestures. For example, the system struggles with statements that need complex cultural or contextual interpretation, like metaphors or regional idioms. Furthermore, the absence of non-manual signals (such as facial expressions and head tilts) in the animations restricts the expressiveness of gestures, which are essential for communicating tone and emotion in sign language. These limitations result from the present dataset's emphasis on conventional BdSL gestures, which do not yet account for regional variances or context-dependent signing styles. User input from native Bangla speakers emphasized the need for better handling of dialect-specific gestures and real-world conversational circumstances. Despite these limitations, the system's HamNoSys-to-SiGML translation and ability to spell out unknown words show that it has the potential to be a fundamental tool in future studies. Addressing these limitations would need increasing the dataset to incorporate idiomatic idioms, cultural subtleties, and non-manual components, as well as using sophisticated NLP algorithms to better understand contextual meaning.

## 5 Conclusion

In conclusion, our study offers noteworthy progress in using visual techniques to create 3D gesture animations in Bangla Sign Language. We offer significant insights and advancements to the industry by coordinating our work with the fundamental concepts of computer graphics and animation. Our method supports the goals of The Visual Computer and associated conferences hosted by the Computer Graphics Society by showcasing the possibilities of combining graphics-oriented methodologies. It is critical to summarize the important results and conclusions of the research while relating them to the scientific challenges stated in the Literature review and introduction. Our research offers a big step toward improving communication accessibility for hearing-impaired Bangla speakers. Our system bridges the gap between deaf and normal people by generating BdSL motions from textual and verbal inputs in novel ways, allowing for seamless communication. With an average accuracy of 97.50% for text input and 94.75% for voice input, our suggested system provides a dependable communication option. Modern HamNoSys to SiGML conversion algorithms have been included, creating new opportunities to improve gesture animation systems. Future studies will also concentrate on multimodal input processing (for example, merging gaze tracking with voice/text inputs) and user customization (for example, avatar personalization). Ethical factors, such as community participation in dataset curation and bias prevention, are critical to guaranteeing the system's inclusiveness and relevance. Addressing intrinsic constraints, such as the need for wider language coverage, Bangla grammar inclusion, and cultural subtleties, is critical for future study efforts. While our SiGML dataset represents progress, more refining is required to capture the nuances of BdSL completely. Future research may focus on expanding the verbal repertoire to accommodate different circumstances, as well as

increasing multimodal input processing, non-manual sign integration, and user customization choices. Emphasizing ethical issues and community participation remains crucial as we imagine a world where people with hearing loss smoothly engage in language and cultural exchanges. In summary, this effort advances the area of automated sign language production, promoting a more inclusive future free of communication hurdles.

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**Data availability** The Zenodo and GitHub repositories contain the source code and datasets created and examined during the present investigation. The datasets are available at <https://doi.org/10.5281/zenodo.12152444>, and the source code is available at <https://gitlab.com/devarifkhan/bdsl-3d-animation>. These resources come with extensive documentation and readme files to make it easier to use them for future studies. It is recommended that researchers make use of these resources to verify findings and investigate new uses.

## Declarations

**Conflict of interest** No conflicting financial or non-financial interests are disclosed by the authors.

**Ethical approval** This article does not contain any studies with human participants or animals performed by any of the authors.

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